

JEE MAIN REVISION MASTER: CHEMISTRY IN EVERYDAY LIFE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 77 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Chemistry In Everyday Life — Core Principles & Governing Laws	64 PYQs	Core Principle
High	Chemistry In Everyday Life — Applied Cases, Reactions & Traps	12 PYQs	Applied Topics
Standard	Chemistry In Everyday Life — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Chemistry In Everyday Life — Core Principles & Governing Laws	$\text{CH}_3(\text{CH}_2)_{16}\text{COO}(\text{CH}_2\text{CH}_2\text{O})_n\text{CH}_2\text{CH}_2\text{OH}, \quad \text{C}_{14}\text{H}_{25}\text{O}_4\text{N}_3\text{S-}\text{Alitame}, \quad (\text{C}_{11}\text{H}_{23}\text{CH}_2\text{OSO}_3^-)\text{Na}^+$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Chemistry In Everyday Life — Core Principles & Governing Laws (Very High Yield — 64 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Chemistry In Everyday Life.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Chemistry In Everyday Life PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II: LIST I LIST II A. Saccharin I. High potency sweetener B. Aspartame II. First artificial sweetening agent C. Alita..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II List I List II A. Tranquilizers I. Anti blood clotting B. Aspirin II. Salvarsan C. Antibiotic III. antidepressant ..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Amongst the following compounds, which one is an antacid?..." [Ans: N/A]

3.2 Chemistry In Everyday Life — Applied Cases, Reactions & Traps (High Yield — 12 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2024 Morning Shift] Q#1 (MCQ):** "Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC II..." **[Ans: B]**
- **[2024 Evening Shift] Q#2 (MCQ):** "Match List - I with List - II. List I (Compound) List II (Use) (A) Carbon tetrachloride (I) Paint remover (B) Methylene chloride (II) Refrig..." **[Ans: N/A]**
- **[2023 Morning Shift] Q#2 (MCQ):** "Given below are two statements: Statement I : Noradrenaline is a neurotransmitter. Statement II : Low level of noradrenaline is not the caus..." **[Ans: N/A]**

3.3 Chemistry In Everyday Life — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- **[2022 Evening Shift] Q#2 (Numerical):** "How many of the following drugs is/are example(s) of broad spectrum antibiotics? Ofloxacin, Penicillin G, Terpineol, Salvarsan...." **[Ans: 1]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2024 (Online) 6th April Morning Shift (Q#1)

Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC III. Antiseptic D. DDT IV. Refrigerants Choose the correct answer from the options given below:

- (A) -II, B-IV, C-I, D-III
- (B) -III, B-I, C-IV, D-II **[Correct]**
- (C) -III, B-II, C-IV, D-I
- (D) -I, B-II, C-III, D-IV

Correct Answer: B

Solution Analysis: The matching for each compound with its respective use is based on their known applications: Iodoform is a compound known for its application as an antiseptic. It has been used historically in medical applications for the treatment of wounds due to its antimicrobial properties. Hence, Iodoform matches with III. Antiseptic. Carbon Tetrachloride was widely used as a fire extinguishing agent due to its property of being a non-flammable liquid that can suffocate fires. However, its use has been significantly reduced in most countries because of its adverse health and environmental impacts. Carbo

Benchmark Problem 2: JEE Main 2023 (Online) 8th April Morning Shift (Q#1)

Match List I with List II: LIST I LIST II A. Saccharin I. High potency sweetener B. Aspartame II. First artificial sweetening agent C. Alitame III. Stable at cooking temperature D. Sucralose IV. Unstable at cooking temperature Choose the correct answer from the options given below:

- (A) -II, B-III, C-IV, D-I
- (B) -II, B-IV, C-I, D-III
- (C) -II, B-IV, C-III, D-I
- (D) -IV, B-III, C-I, D-II

Correct Answer: None

Solution Analysis: A. Saccharin is first popular artificial sweetening agent. It is 550 times as sweet as cane sugar. It is used as intake by diabetic person. B. Aspartame is 100 times as sweet as cane sugar. It is used in cold foods and soft drinks because it is unstable at cooking temperature. C. Alitame is 2000 times as sweet compared to cane sugar. It has excellent stability at high temperature. So, can be used in cooking and baking. D. Sucralose is 600 times as sweet as cane sugar. It is heat stable and finds its use in baked goods.

Benchmark Problem 3: JEE Main 2022 (Online) 26th July Evening Shift (Q#2)

How many of the following drugs is/are example(s) of broad spectrum antibiotics? Ofloxacin, Penicillin G, Terpineol, Salvarsan.

Correct Answer: 1

Solution Analysis: Ofloxacin is the only broad-spectrum antibiotic given in the question Penicillin - G is a narrow-spectrum antibiotic. Salvarsan is mainly active against spirochete, a bacteria that causes syphilis Terpineol is an antiseptic.